

COOCH BEHAR GOVERNMENT ENGINEERING COLLEGE

Department of Computer Science & Engineering

ASSIGNMENT-III [CSE 3RD SEMESTER]

IT WORKSHOP LAB [Python Programming]

1. A list is a palindrome if it reads the same forwards and backwards. For instance [], [7], [8,11,8] and [19,3,44,44,3,19] are palindromes, while [3,18,4] and [23,14,3,14,3,23] are not. Write a recursive function to check if a list is a palindrome.
2. A positive integer n is said to be perfect if the sum of the factors of n , other than n itself, add up to n . For instance 6 is perfect since the factors of 6 are {1,2,3,6} and $1+2+3=6$. Likewise, 28 is perfect because the factors of 28 are {1,2,4,7,14,28} and $1+2+4+7+14=28$.

Write a Python function perfect(n) that takes a positive integer argument and returns True if the integer is perfect, and False otherwise.

3. Write a Python function sublist($l1,l2$) that takes two **sorted** lists as arguments and returns True if the first list is a sublist of the second list, and returns False otherwise. A sublist of a list is a segment consisting of contiguous values, without a gap. Thus, [2,3,4] is a sublist of [2,2,3,4,5], but [2,2,4] and [2,4,5] are not.
4. A list is a non-decreasing if each element is at least as big as the preceding one. For instance [], [7], [8,8,11] and [3,19,44,44,63,89] are non-decreasing, while [3,18,4] and [23,14,3,14,3,23] are not. Write a recursive function to check if a list is non-decreasing or not.
5. A positive integer n is a sum of squares if $n = i^2 + j^2$ for integers i,j such that $i \geq 1$ and $j \geq 1$. For instance, 10 is a sum of squares because $10 = 1^2 + 3^2$, and so is 25 ($3^2 + 4^2$). On the other hand, 11 and 3 are not sums of squares.

Write a Python function sumofsquares(n) that takes a positive integer argument and returns True if the integer is a sum of squares, and False otherwise.

6. Write a Python function subsequence($l1,l2$) that takes two **sorted** lists as arguments and returns True if the first list is a subsequence of the second list, and returns False otherwise.

A subsequence of a list is obtained by dropping some values. Thus, [2,3,4] and [2,2,5] are subsequences of [2,2,3,4,5], but [2,4,4] and [2,4,3] are not.

7. A positive integer n is a sum of three squares if $n = i^2 + j^2 + k^2$ for integers i,j,k such that $i \geq 1, j \geq 1$ and $k \geq 1$. For instance, 29 is a sum of three squares because $10 = 2^2 + 3^2 + 4^2$, and so is 6 ($1^2 + 1^2 + 2^2$). On the other hand, 16 and 20 are not sums of three squares.

Write a Python function `sumof3squares(n)` that takes a positive integer argument and returns `True` if the integer is a sum of three squares, and `False` otherwise.

8. Write a Python function `intersect(l1,l2)` that takes two **sorted** lists as arguments and returns the list of all elements common to both `l1` and `l2` in the same order that they appear in the two lists. If the same element occurs more than once in both lists, it should appear in the output exactly once.

Thus, `intersect([2,2,4],[1,2,2,3,4])` should return `[2, 4]` while `intersect([1,2,3],[4,5,6])` should return `[]`.